PROBLEM SHEET 2

1. a. $H: P \to Q$
$H: Q \to R$
$C: P \to R$

P	Q	R	$P \to Q$	$Q \to R$	$P \to R$
0	0	0	1	1	1
0	0	1	1	1	1
0	1	0	1	0	1
0	1	1	1	1	1
1	0	0	0	1	0
1	0	1	0	1	1
1	1	0	1	0	0
1	1	1	1	1	1

valid

b. $H: \sim P \to Q$
$H: \sim(Q \wedge \sim R)$
$H_0: \sim R$
$C: \sim P$

valid

P	Q	R	$\sim P$	$\sim R$	$\sim P \to Q$	$\sim(Q \wedge \sim R)$
0	0	0	1	1	0	0 1
0	0	1	1	0	0	0 1
0	1	0	1	1	1	0 1
0	1	1	1	0	1	0 1
1	0	0	0	1	1	0 1
1	0	1	0	0	1	0 1
1	1	0	0	1	1	0 1
1	1	1	0	0	1	0 1

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Disjunctive Airtonc A B ictuse

2. a. $A \wedge B \wedge ((A \leftrightarrow B) \rightarrow (C \vee D)) \Rightarrow D \vee C$

~~$\times$ $A \wedge B$ $((A \rightarrow B) \wedge (B \rightarrow A))$ $\vee ((A \wedge B) \rightarrow (C \vee D))$~~

~~$\times$ $A \wedge B$ $(A \wedge \neg B) \wedge (B \wedge A) \vee (C \vee D) \Rightarrow D \vee C$~~

$(A \wedge B) \vee (A' \wedge B')$

$(A \leftrightarrow B) \wedge ((A \leftrightarrow B) \rightarrow (C \vee D)) \Rightarrow D \vee C$

$(C \vee D) \rightarrow D \vee C.$

b. $(P \rightarrow Q) \wedge (\neg Q \vee R) \wedge \neg R \wedge \neg(\neg P \wedge S) \Rightarrow \neg S$

$(P \rightarrow Q) \wedge (Q \rightarrow R) \wedge \neg R \wedge \neg(S \rightarrow P) \Rightarrow \neg S$

$(P \rightarrow R) \wedge \neg R \wedge (S \rightarrow P) \Rightarrow \neg S$

$\neg P \wedge (S \rightarrow P) \Rightarrow \neg S$

$\neg S \Rightarrow \neg S.$

c. $\neg J \rightarrow (M \vee N) \wedge ((H \vee G) \rightarrow \neg J) \wedge (H \vee G) \Rightarrow M \vee N$

$[\neg J \rightarrow (M \vee N)] \wedge \neg J \Rightarrow M \vee N$

$M \vee N \Rightarrow M \vee N.$

d. $(P \vee Q) \wedge (P \rightarrow R) \wedge (Q \rightarrow S) \Rightarrow S \vee R$

$(\neg P \rightarrow S) \wedge (P \rightarrow R)$

$(P \vee S) \wedge (\neg P \vee R)$

$(S \vee R)$

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$\sim R = 1 \quad R = 0 \quad P \to R \;\; 1$

$0 \quad 0$

3. a. $(P \to Q) \wedge (Q \to R) \wedge \sim(P \wedge R) \wedge (P \vee R) \wedge \sim R$

$(P \to R) \wedge (\sim P \vee \sim R) \wedge (P \vee R) \wedge \sim R$

$\sim P \wedge (\sim R \vee R)$

$\sim P$

F.

b. $(S \to \sim Q) \wedge (S \vee R) \wedge \sim R \wedge (\sim R \leftrightarrow Q) \wedge \sim P$

$(S \to \sim Q) \wedge (\sim S \to R) \wedge \sim R \wedge (\sim R \leftrightarrow Q) \wedge P$

$(S \to \sim Q) \wedge (\sim R \to S) \wedge \sim R \wedge (\sim R \leftrightarrow Q) \wedge P$

$(\sim R \to \sim Q) \wedge \sim R \wedge (\sim R \leftrightarrow Q) \wedge P$

$\sim Q \wedge P \wedge (\sim R \leftrightarrow Q)$

$\sim Q \wedge P \wedge (\sim R \to Q) \wedge (Q \to \sim R) \wedge P$

$R \wedge P \wedge (Q \to \sim R) \wedge P$

$P \wedge \sim Q$

F.

c. $[\sim(P \to Q) \to \sim(R \vee S)] \wedge [(Q \to P) \vee \sim R] \wedge R$

$\wedge \sim(P \leftrightarrow Q)$

$\wedge \sim(P \leftrightarrow Q)$

$\wedge \sim(P \to Q) \wedge \sim(Q \to P)$

$\sim(R \vee S) \wedge [(Q \to P) \vee \sim R] \wedge R$

$\wedge \sim(Q \to P)$

$\sim R \wedge \sim S \wedge [(Q \to P) \vee \sim R] \wedge R \wedge$

$\sim(Q \to P)$

$F \wedge$

d $P \to (Q \to R) \wedge (\sim S \vee P) \wedge Q \wedge \sim(S \to R)$

$P \to (Q \to R) \wedge (\sim S \vee P) \wedge Q \wedge S \wedge \sim R$

$P \to (Q \to R) \wedge (S \to P) \wedge Q \wedge S \wedge \sim R$

$P \to (Q \to R) \wedge P \wedge Q \wedge \sim R$

$Q \to R \wedge Q \wedge \sim R$

$R \wedge \sim R$

F

4. a. S → The system is in malliuser state
O → It is operating normally
K → The kernal is functioning
I → The system is in interrupt mode

H_1: $O \to S$
H_2: $O \to K$
H_3: $\sim K \vee I$
H_4: $\sim S \to I$
H_5: $\sim I$

$O \to S \wedge O \to K \wedge K \to I \wedge \sim S \to I \wedge \sim I$

$O \to S \wedge O \to I \wedge \wedge \sim S \to I \wedge \sim I$

$O \to S \wedge \sim O \wedge \sim S \to I \wedge \sim I$ (add $\sim I$)

$O \to S \wedge \sim O \wedge S$

$O \sim O \wedge S$

F

b. R → Routers can send packets to the edge system

S → Routers supports the new address space

L → Latest software release is installed

H: R → S
H: L → S
H: L → R
H: ~S

$R \to S \land L \to S \land L \to R \land \sim S$

$L \to S \land \sim S$

$\sim L$

True

L : File system is now locked
M : New messages will be queued
S : System is functioning normally
B : Messages sent to message buffer.

H: L → M
H: L → S
H: ~M → B
H: L → B
H: ~B

$L \to M \land L \to S \land \sim M \to B$

$L \to R \land \sim B$

$L \to M \land L \to S \land \sim M \to B \land \sim L$

$L \to M \land (L \to S) \land S \to L \land \sim M \land \sim L$

$L \to M \land L \to S \land \sim S \land \sim M \to B \land \sim L$

$\sim L \to 1$ Then $L \to S = 1$

$\sim S \to 1$ Then $L \to M = 1$

$\sim M \to B$

$L \to 0$

$S \to M \land L \to S \land \sim M \to B \land \sim L$

$S \to M \land L \to S \land \sim B \to M \land \sim L$

$S \to \sim B \land L \to S$

$L \to \sim B \land \sim L$

T

d. M - Miranda doesnt take the course in DMath
G -> She will not Graduate
Q -> Not qualified for the job.
R -> Miranda sends the book.

H: $M \to G$ H: $G \to Q$ H: $R \to \sim Q$ H: $M \wedge R$.

$M \to G \wedge G \to Q \wedge R \to \sim Q \wedge M \wedge R$

$M \to Q \wedge R \to \sim Q \wedge M \wedge R$

$Q \wedge \sim Q$

F

$M \to Q \wedge R \to \sim Q \wedge M \wedge R$

$Q \wedge R \to \sim Q \wedge R$

$\Rightarrow R \wedge R \Rightarrow F$

Inconsistent.

I.e.

B -> play Basketball
F -> Play Football
S -> Today is Saturday
R -> Tomorrow is FRIDAY

H: $B \wedge F$

H: $\sim S \to (B \wedge F)$

H: $(S \vee R) \to \sim F$

C: $\sim F$

$B \wedge \sim S \to (B \wedge F) \wedge \sim(S \vee R)$

$B \wedge \sim S \to (B \wedge F) \wedge \sim S \wedge \sim R$

$B \wedge B \wedge F \wedge \sim R \Rightarrow \sim F$

$B \wedge F \wedge \sim S \to (B \wedge F) \wedge (S \vee R) \Rightarrow \sim F$

$B \wedge \sim(S \vee R) \wedge \sim S \to (B \wedge F) \Rightarrow \sim F$

$B \wedge \sim S \wedge \sim R \wedge \sim S \to (B \wedge F) \Rightarrow \sim F$

$B \wedge B \wedge F \wedge \sim R \Rightarrow \sim F$

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$C \vee R$ $\quad$ $C \vee R \wedge \sim R$

$R \to M$ $\quad$ $(\sim C \to R) \wedge \sim R$

$\sim M$ $\quad$ C

b. $C \to$ Frank bought a PC

$V \to$ " " " " $\vee$ C R.

$M \to$ Likes to watch movies at Home

H: $C \vee V$ $\quad$ $(C \vee V) \wedge (V \to M) \wedge \sim M \to$

H: $V \to M$ $\quad$ $(\sim V \to C) \wedge \sim V \to C$

H: $\sim M$ $\quad$ $C \to C$

C: C

c. $M \to$ The Band can play rock music

$T \to$ Replacements were not delivered on time

$C \to$ New years party cancelled

$A \to$ Alicia would have been angry.

$R \to$ Refunds would have had to made

H: $(\sim M \vee T) \to (C \wedge A)$

H: $C \to R$

H: $\sim R$

C: M.

$(\sim M \vee T) \to (C \wedge A) \wedge \sim C \to M$

$(M \wedge \sim T) \vee (C \wedge A) \wedge \sim C \to M$

$M \wedge \sim T \wedge \sim C \to M$

$M \to M$

$\sim(\sim M \vee T) \vee (C \wedge A) \wedge (\sim C \vee A)$

$M \wedge \sim T$

M

$A \to (B \wedge C)$

$A \to B \wedge A \to C$

3c. $\sim(P \to Q) \to \sim(R \vee S) \wedge (Q \to P) \vee \sim R \wedge R$

$\to \sim(P \leftrightarrow Q)$

$\sim(P \to Q) \to (\sim R \vee S) \wedge ((Q \to P) \vee \sim R) \wedge R$

$\wedge \sim(P \wedge Q) \wedge \sim(\sim P \wedge \sim Q)$

$\sim(P \to Q) \to (\sim R \wedge \sim S) \wedge ((Q \to P) \vee \sim R) \wedge R$

$\wedge \sim(P \wedge Q) \wedge \sim(\sim P \wedge \sim Q)$

$\sim(P \to Q) \to \sim R \wedge \sim(P \to Q) \to \sim S$

$(P \to Q) \wedge \sim(P \to Q) \to \sim S \wedge \sim(P \leftrightarrow Q)$

$\wedge ((Q \to P) \vee \sim R)$

Add R again.

$(P \to Q) \wedge \sim(P \leftrightarrow Q) \wedge ((Q \to P) \vee \sim R) \wedge R$

$(P \to Q) \wedge \sim(P \leftrightarrow Q) \wedge (R \to (Q \to P)) \wedge R$

$(P \to Q) \wedge (Q \to P) \wedge \sim(P \leftrightarrow Q)$

$(P \leftrightarrow Q) \wedge \sim(P \leftrightarrow Q)$

F.

6. b. No rabbit knows calculus.

~~P(x) → x knows calculus.~~

~~∃x P(x) → There is a~~

P(x) : For all x, they dont know calculus.

$\forall x . P(\text{rabbit})$

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b. For every rabbit they dont know Calculus

$\forall x\, P(x)$ where

$P(x)$ is x dosent know Calculus.

$R(x)$ is x is a rabbit

$\forall x\, (R(x) \rightarrow P(x))$.

c. There is a student in this class and that student took a discrete math class and dosent know Python

$S(x)$ = x is a student

$Q(x)$ = x took a discrete math class

$P(x)$ = x dosent know Python

$\forall x\, (S(x) \wedge Q(x) \wedge P(x))$.

d. $S(x)$ = x is a student

~~$D(x, y)$ = x took y department~~

$D(y, z)$ = course y is in z department

~~$\forall x \exists y \forall z\, (S(x) \rightarrow$~~

$D(x, y, z)$ = x took y course in z department

$\forall x \exists y \forall z\, (S(x) \rightarrow D(x, y, z))$

B(x) = x is ... Building exist ...
W(x,y) = Every room in x is coloured white
e. There exists a building in USA ...

e. There exists a building on the campus in at least one clg of USA and in that building every room is painted white.

B(x) -> x is a building on the campus of colleges of USA

R(x,y) W(y) -> ~~Every room is painted white in building y.~~ y is a room in building x

W R(z) -> That Room z is painted white.

$\exists x \forall y \forall z \, (B(x) \wedge R(x,y) \rightarrow W(z))$

f. Everyone in your class with an internet connection has chatted over the internet with at least one other student in your class.

S(x) -> x is a stud of your class

I(x) = x has internet connection.

C(x,y) = x has chatted with y.

$\forall x \exists y \, ((S(x) \wedge I(x)) \rightarrow C(x,y))$

g. Sanjay has chatted with everyone except Joseph.

P(x,y) = x has chatted with y ~~except~~

~~P(Sanjay, Joseph)~~

$\forall y (y \neq \text{Joseph} \Rightarrow P(\text{Sanjay}, y))$

h. $H(x)$ = x knows Hindi $S(x)$ x is a student in this class
$F(x)$ = x knows French

$\forall x (S(x) \rightarrow (H(x) \wedge F(x)))$

i. Not all college grads are smart

$\forall x (C(x) \wedge \sim S(x))$

$C(x) \Rightarrow$ x is a college grad.
$S(x) \rightarrow$ x is smart.

j. People who play golf are kind to animals.

$G(x) \rightarrow$ x plays golf
$K(x) \rightarrow$ x is kind to animals

$\forall x (G(x) \rightarrow K(x))$.

K. Everyone knows someone who plays golf.

$\forall x \exists y (P(x,y) \wedge G(y))$

$P(x,y)$ = x knows y
$G(y)$ = y plays golf

L. No bird eats only worms.

$B(x) \Rightarrow$ x is a bird.
$W(x) \Rightarrow$ x eats only worms.

$\forall x (B(x) \rightarrow \sim W(x))$

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m. Every commille member is rich and famous.

$C(x) \to$ x is a commillemand

$R(x) \to$ x is rich

$F(x) \to$ x is famous.

$\forall x\,(C(x) \to (R(x) \wedge F(x)))$.

7.

a. $\forall x\,(P(x) \wedge R(x)) \to (\forall x)\,P(x) \wedge Q(x)$

free

b. $\forall(x)\,(P(x) \wedge (\exists x)\,Q(x)) \vee [\forall x\,P(x) \to Q(x)]$

free

c. $\cancel{(\exists x)\,P(x) \wedge \forall x\,[P(x) \wedge Q(x)] \to \exists x\,Q(x)}$

S(x)

for ∞

8.

a. $(\forall x)(\forall y)\,[P(x,y) \to Q(x,y)] \wedge \sim Q(a,b) \Rightarrow \sim P(a,b)$

Universal ~~modus ponens~~ ~~Distribution~~ Instantiation

Modus Tolens.

b. $\sim[\exists x\,P(x) \wedge Q(x)] \Rightarrow \exists x\,P(x) \to \sim Q(a)$

From Existential instantiation.

$\sim[P(c) \wedge Q(c)]$ Demorgan

$\sim P(c) \vee \sim Q(c)$ Conditional

$P(c) \to \sim Q(c)$ Existential

$\exists x\,P(x) \to \sim Q(x)$ generalization

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c. $\exists x\, P(x) \wedge \forall x\, [P(x) \rightarrow Q(x)] \Rightarrow \exists x\, Q(x)$

Universal instantiation and Existential instantiation

$P(a) \wedge P(a) \rightarrow Q(a)$ Modus Ponens.

$Q(a)$ Existential generalization

$\exists x\, Q(x)$.

9. ~~$\forall x\, [P(x) \rightarrow Q(x)]$ $\Rightarrow \forall x\, P(x) \rightarrow \forall x\, Q(x)$~~

~~$P(a) \rightarrow Q(a)$ is true. $\Rightarrow$ Universal Instantiation.~~

10. a. Jack owns a dog. Every dog owner is a animal lover. No animal lover kills an animal. Either Jack or Curiosity killed the cat, who is named Tuna.

$D(x) \Rightarrow$ x owns a dog.

$L(x) \Rightarrow$ x is a animal lover.

$K(x) \Rightarrow$ x did not killed a animal.

$D(x)$ $\qquad$ $D(Jack)$.

$\forall x\, (D(x) \rightarrow L(x))$.

$\forall x\, (L(x) \rightarrow K(x))$.

$\neg K(Jack) \vee \neg K(Tuna)$.

$D(Jack) \wedge \forall x\, (D(x) \rightarrow L(x)) \wedge \forall x\, (L(x) \rightarrow K(x))$
$\wedge\, (\neg K(Jack) \vee \neg K(Tuna))$

U. I.

$D(Jack) \wedge D(Jack) \rightarrow L(Jack) \wedge (L(Jack) \rightarrow K(Jack))$

$\wedge \neg \ldots$

$D(Jack) \wedge D(Jack) \rightarrow K(Jack)$.

$K(Jack)$

If $K(Jack)$ is true then Jack didnot Kill ~~[illegible]~~ Tuna or who ever the Shit that was.

b. John likes all kinds of food. Apples are food Chicken is food. Anything anything anyone eats and is not killed by is food. Bill eats peanuts and is still alive. Sue eats everything Bill eats. Prove John likes peanuts.

~~F(x) is x is food.~~

$A(x) \Rightarrow$ x is Alive

$E(x,y) =$ x ate y.

$F(y) \Rightarrow$ y is food.

$L(x,y) =$ x likes y.

Given $\forall y\, L(John, y)$.

$\forall x \forall y\, (A(x) \wedge E(x,y) \rightarrow F(y))$.

$E(Bill, peanut) \wedge A(Bill)$

UI

$\forall y\, L(John, y) \wedge F(peanuts)$.

U. Instorsiation

$L(John, Peanuts)$.

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